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Data Science

AI Enabled Smart Road Safety Violation Detection And Monitoring System Using Neural Vision

Group No. 5

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Abstract

Traffic rule enforcement is often slow, error-prone, and labor-intensive. This project introduces a smart AI system using YOLOv8 and ANPR to detect violations like helmet-less riding, triple riding, seat belt absence, red light jumping, wrong-side driving, and mobile usage in real time. Each violation is logged with vehicle details, timestamp, and location, and fines are automatically generated and sent via email or SMS. By combining live video feeds, AI detection, and automated reporting, this system ensures faster, accurate, and consistent traffic law enforcement, making roads safer and smarter.

Sustainable Development Goals(SDG) mapped

SDG mapped to project: 11

SDG 11: Sustainable Cities and Communities

Mapping: Automated traffic monitoring plays a crucial role in creating safer roads and promoting better traffic discipline. By detecting violations in real time, it helps reduce accidents and ensures compliance with traffic rules. This contributes to more organized and livable cities by streamlining traffic flow and reducing congestion. Moreover, it encourages sustainable urban mobility and fosters public trust in smarter, technology-driven traffic management systems.

Introduction

- Road accidents are a major concern due to frequent traffic violations such as riding without helmets, triple-seat riding, and jumping red lights. Despite awareness campaigns, these unsafe behaviors continue to put lives at serious risk.
- Current traffic monitoring relies heavily on manual enforcement by traffic police, which is limited by human capacity, visibility, and availability.
- Manual monitoring cannot effectively cover all areas, especially in high-traffic zones, resulting in many violations going unnoticed and delayed action.
- The inefficiency of existing systems reduces the overall effectiveness of road safety measures, highlighting the need for smarter, automated solutions

Motivation

- Increasing road accidents and fatalities caused by preventable violations such as not wearing helmets or triple riding strongly motivate the development of automated enforcement systems.
- Automating traffic violation detection using AI ensures faster, unbiased, and scalable enforcement. This approach addresses issues such as unidentified violators, delayed fine generation, and repeated offenses, ultimately promoting safer roads and disciplined traffic behavior

Objectives

1. To integrate mobile camera live video capture with an external processing system for real-time AI-powered violation detection and data streaming.

2. To apply AI models for automatic detection of traffic violations such as helmet-less riding, triple seat riding, seat belt absence, red light jumping, wrong-side driving, and mobile phone usage, smoking while driving.

3. To extract vehicle number plates using ANPR techniques and log each violation with its type, timestamp, and location metadata and manage them by marking their status followed by automated email notifications to violators.

4. To enable automated generation of e-challans for detected violations, ensuring accurate fine calculation and streamlined digital penalty issuance without manual intervention.

Literature Review

The literature review is presented in the sequence illustrated in the figure: Live Video Capture & Streaming, AI-Based Violation Detection, Vehicle Identification & Logging, and Automated E-Challan Generation.

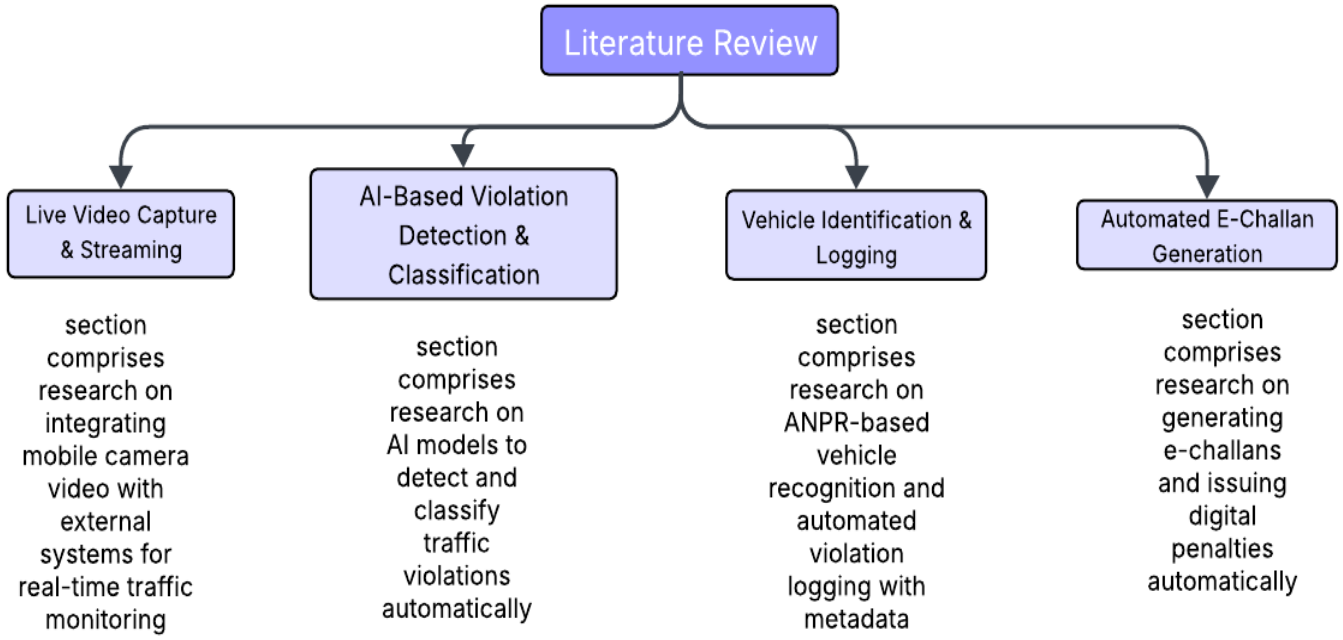


Figure 1: Categorized literature review

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	AI-Based Helmet Violation Detection for Traffic Management System[1]	Yahia Said, Yahya Alassaf, Refka Ghodhbani, Yazan Ahmad Alsariera	2025	Adapted PerspectiveNet with EfficientNetv2 backbone; trained on India Driving Dataset (IDD) for helmet detection	Focused only on helmet detection; may struggle with occlusion, multiple riders, or real-world varied conditions
2	Automatic Two-Wheeler Rider Identification and Triple-Riding Detection[2]	Uttam U. Deshpande, Supriya Shanbhag, Rudragoud Patil	2025	This paper presents a ResNet18-based DetectNet_v2 for two-wheeler detection; YOLOv8 on Darknet for triple-riding and license plate detection	Dataset has minimal triple-riding images; may not generalize to all CCTV angles and real-world scenarios
3	Wrong-Way Driving Detection with YOLOv9 and CNN[3]	Kiranmaie Pa, Abhisatwika Reddy Chada, Ilaiah Kavati	2025	YOLOv9 for vehicle detection; CNN for vehicle direction classification; trained on annotated images of correct and incorrect driving	Limited to wrong-way detection; may fail in complex intersections or occluded scenarios; dataset may not cover all traffic conditions
4	Real-time license plate detection for non-helmeted motorcyclist using YOLO[4]	Yonten Jamtsho, Panomkhawn Riyamongkol, Rattapoom Waranusast	2020	A single convolutional neural network (YOLOv2) is used for real-time detection of license plates for non-helmeted motorcyclists, and a centroid tracking method is used to reduce false positives from helmeted bikers leaving the frame.	The system's accuracy is affected by the absence of a license plate on some motorbikes and it relies on rear-position plates, as front LPs are not available for Thai motorcyclists.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
5	Automated Detection of Driver and Passenger Without Seat Belt using YOLOv8[5]	Sutikno, Aris Sugiharto, Retno Kusumaningrum	2023	The system is divided into three subsystems—windshield detection, passenger classification, and seat belt classification—all of which are implemented using different YOLOv8 models.	A key limitation is the absence of a license plate detection subsystem, which is noted as a component for future research to enable the identification of vehicle owners.
6	Image-Based Seat Belt Fastness Detection Using Deep Learning[6]	Rupal A. Kapdi, Pimal Khanpara, Rohan Modi, and Manish Gupta	2022	Proposes an approach using a CNN-based model with MobileNet V2 as the backbone and transfer learning for binary seat belt classification, complemented by a number plate recognition module that uses a KNN classifier.	The dataset is highly imbalanced, leading to a zero recall and F1-score for the "without seat belt" class before threshold tuning. The system struggles with images that are too bright or dark, and with occlusions from objects or clothing that blend with the seat belt.
7	A Comprehensive Analysis of Real-Time Car Safety Belt Detection Using the YOLOv7 Algorithm[7]	Lwando Nkuzo, Sibiya, Markus, Elisha Didam Markus	2023	A real-time system uses a YOLOv7 object detection model, trained on a custom dataset annotated with "buckled" and "unbuckled" classes, to detect seat belt status by focusing on the visibility of the pillar loop.	The system's performance is limited by challenging environmental conditions such as high-light intensity, occlusions of the pillar loop, and blurry images, and it was not evaluated in night-time scenes.
8	DashCop: Automated E-ticket Generation for Two-Wheeler Traffic Violations Using Dashcam Videos[8]	Deepti Rawat, Keshav Gupta, Aryamaan Basu Roy, and Ravi Kiran Sarvadevabhatla	2025	The system uses a modified YOLOv8 with a novel Segmentation and Cross-Association (SAC) module and a custom tracker to detect violations and a separate module for automated license plate recognition, enabling end-to-end E-ticket generation.	The system struggles with significant glare, motion blur, and occlusions, which can lead to false negatives in detection, and license plate recognition performance decreases for vehicles approaching from the opposite lane.

Research Gap(Limitations of existing systems)

- **Low-Light and Weather Challenges**: Most current traffic violation systems struggle to maintain detection accuracy in poor lighting, nighttime conditions, or adverse weather such as rain and fog.
- **Occlusion and Motion Blur**: Vehicles or objects blocking the camera view, as well as fast-moving traffic, lead to missed or incorrect violation detection.
- **License Plate Recognition Issues**: ANPR systems often fail to read dirty, damaged, or regionally varying license plates, resulting in incomplete offender identification.
- **Generic Reporting** – Many platforms generate standard violation reports without actionable insights or analytics for traffic officers.
- **Delayed or Inaccurate e-Challan Generation**: Errors or processing delays in automated challan issuance occur due to incomplete metadata or high traffic volume.
- **Scalability and Adaptability Limitations**: Existing systems often cannot handle multiple camera feeds efficiently and may not adapt well to different traffic densities or regional variations.
- **Partial Integration of Advanced AI**: While deep learning models like YOLOv8 improve multi-class detection, they are not widely implemented or optimized, leaving room for improvement in real-time accuracy and robustness.

Problem Definition

Road accidents are a major cause of fatalities worldwide due to unsafe driving practices. Common violations include riding without helmets, triple riding, mobile phone usage, and not wearing seatbelts. Manual monitoring by traffic police is inefficient, error-prone, and cannot cover all areas effectively. There is a need for an automated AI-based system for real-time traffic violation detection. Such a system can identify offenders, log violations, generate e-challans, and improve overall road safety.

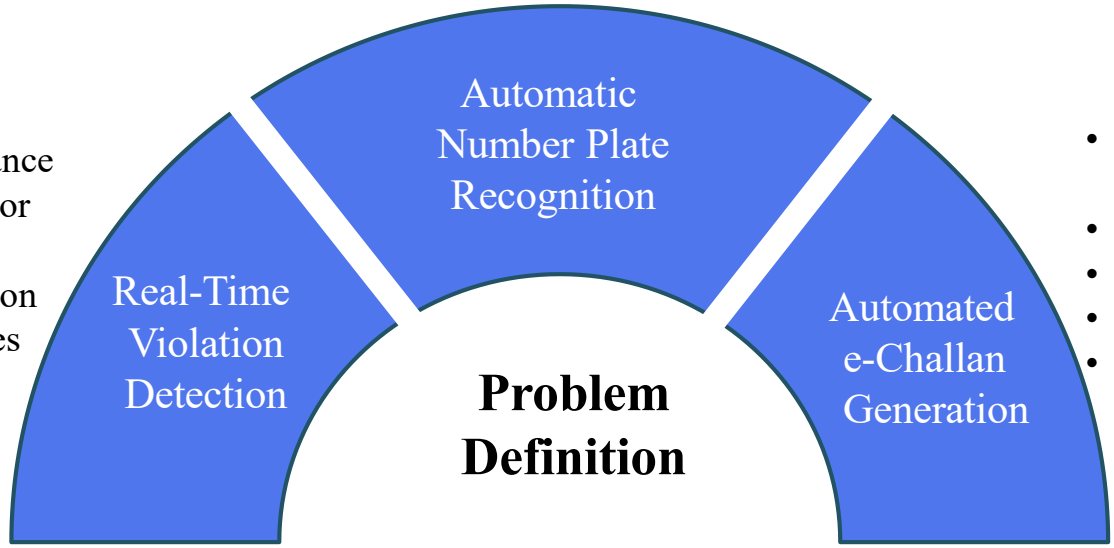
Key points to address:

1. Real-Time Violation Detection
2. Automatic Number Plate Recognition
3. Live Camera Integration and Video Upload
4. Violation Metadata Logging
5. Automated e-Challan Generation
6. Traffic Officer Dashboard
7. Reports and Analytics

Problem Definition

- Dirty or occluded plates
- Low-resolution cameras
- Plates from different regions
- Fast-moving vehicles reduce OCR accuracy
- Limited training data for uncommon plate formats

- Poor lighting performance
- Occlusion by vehicles or objects
- Low frame rate detection
- Crowded traffic reduces accuracy
- Motion blur in fast-moving vehicles



- Accuracy issues in detected violations
- Delays in processing
- Incomplete violation metadata
- Wrong notifications due to errors
- Scalability issues with high traffic volume

Figure 2: Problem Definition Diagram

Scope

- Detects traffic violations in real-time including no helmets, triple riding, seatbelt absence, and mobile phone usage, improving enforcement efficiency
- Performs Automatic Number Plate Recognition to accurately identify vehicles and link violations to owners
- Supports integration of live camera feeds and video uploads for flexible monitoring across multiple locations
- Generates automated e-challans and sends instant notifications to offenders, reducing manual intervention
- Provides interactive dashboards for traffic officers with login, access control, real-time monitoring, and violation management
- Offers analytics and insights for violation trends, high-risk areas, and data-driven decision making to enhance road safety

Technological Stack

Front-end:

- React.js/HTML5/CSS3/JavaScript**:** For building responsive web interfaces, ensuring cross- device compatibility

Back-end:

- Django 4.x / Django REST Framework: API handling, user authentication, and backend processing.
- PostgreSQL / MySQL: Stores violation logs, vehicle details, timestamps, and status.

Models:

- YOLOv8 (PyTorch): Multi-class traffic violation detection.
- ANPR Model (Tesseract OCR / EasyOCR): Reads vehicle license plates

Datasets:

- Kaggle Traffic Violation Dataset: Pre-collected videos/images annotated for multiple violation classes.
- Custom Data Augmentation: Flip, rotate, scale, brightness adjustment to improve model robustness.

Technological Stack

Algorithm:

- CNN-based Algorithm: Specialized classification or additional violation types
- Fine Calculation Algorithm: Computes penalties based on type & frequency of violation

Libraries:

- OpenCV / NumPy: Video frame processing and image manipulation.
- SMTP / SendGrid API: Automated email notifications to violators.
- CUDA / cuDNN: GPU acceleration for real-time inference with YOLOv8.

Analytics:

- Violation Statistics : Counts and categorizes violations by type, location, and time of day using Python .
- Dashboard Visualization: Interactive dashboards with Plotly, Matplotlib, Seaborn

Proposed system architecture

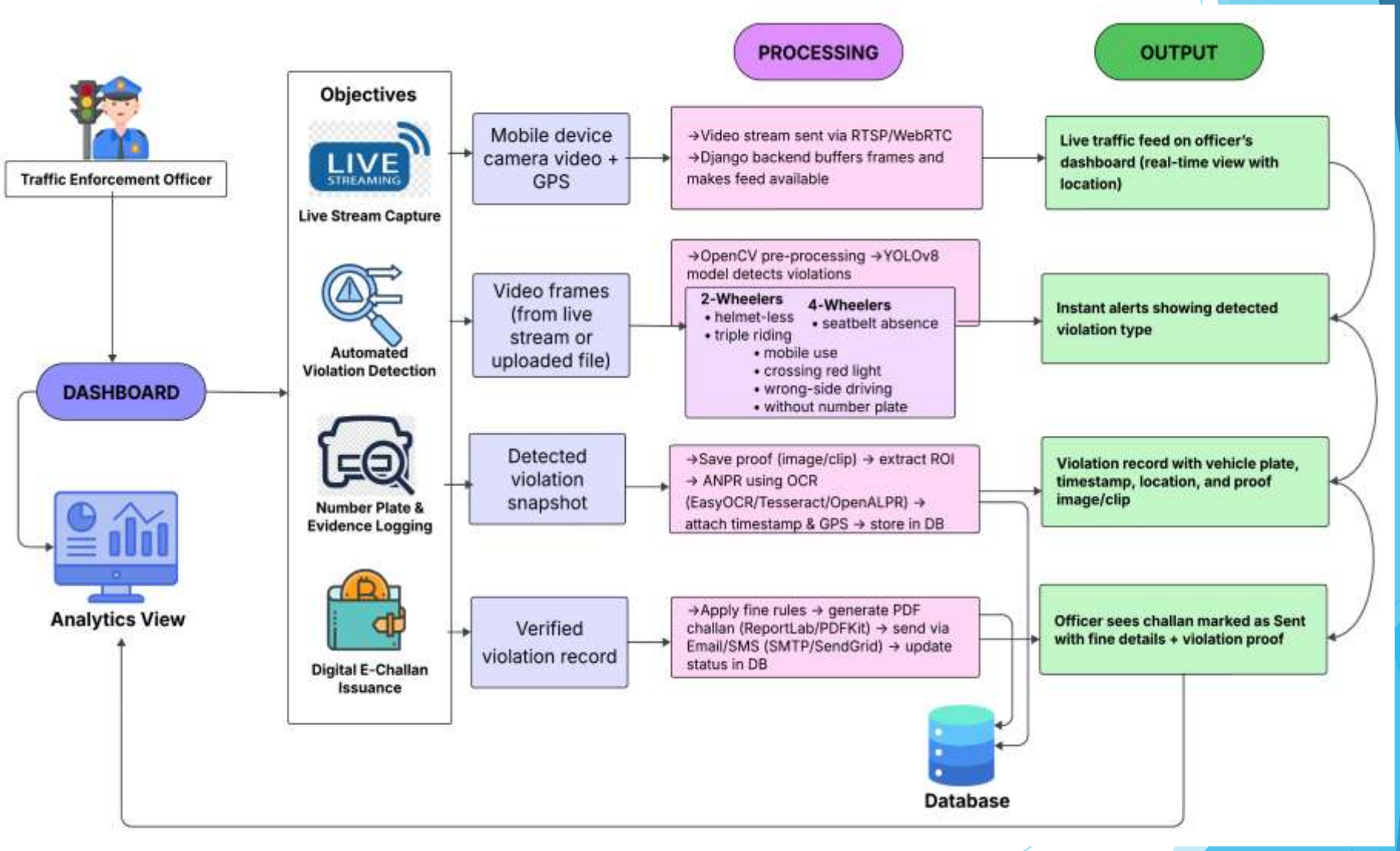


Figure 3: Proposed System Architecture

Prototype Design Demonstration

Not Applicable for Review 1

Implementation Status

Helmetless Riding & Number Plate Detection: Built models using YOLO to automatically detect riders without helmets and capture vehicle number plates for monitoring and enforcement.

Implementation Status

Triple-Seating Violation: Developed a YOLO-based model to identify vehicles carrying more than the allowed number of passengers.

Implementation Status

Seatbelt, Smoking, and Phone Use Violations: Implemented YOLO models to detect drivers not wearing seatbelts, smoking, or using mobile phones while driving, enhancing overall road safety monitoring.

Review Suggestions (Given in Last meeting)

Not Applicable for Review 1

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Thank You...!!